Article

Dynamic Prototype Routing for Lightweight Image Super-Resolution

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Featured Application

Confidence-aware content routing for efficient single-image super-resolution on resource-constrained platforms such as mobile and edge devices.

Abstract

Lightweight single image super-resolution (SR) must balance reconstruction quality with model cost on resource-constrained platforms. Content-aware routing improves this balance by grouping semantically related tokens, but the Token Aggregation Block (TAB) of CATANet uses cross-batch exponential-moving-average centers, hard assignment labels, and a coupled prototype update/confirmation step. We propose *Dynamic Prototype Routing* (DPR), a drop-in TAB replacement that keeps the CATANet backbone and tensor interface unchanged. DPR forms per-image prototypes from the current tokens, refines them with a learnable query bank, matches tokens with a bounded learnable temperature, and carries assignment confidence into token ordering, global gating, and a soft fallback; a balancing loss discourages prototype-slot collapse. Trained on DIV2K and evaluated on five standard benchmarks at $\times 2/\times 3/\times 4$, DPRNet stays in the lightweight regime and matches CATANet-level accuracy, with small gains on most benchmark cells (e.g. 26.89 dB on Urban100 at $\times 4$ with 52.14 G Multi-Adds). It uses fewer Multi-Adds than the lightweight Transformer baselines SwinIR-light and SRFormer-light under the same input convention, while providing interpretable routing diagnostics in which confidence remains soft and prototype usage becomes more balanced.

Keywords: image super-resolution; lightweight super-resolution; content-aware routing; dynamic prototype routing; self-attention; vision transformer; deep learning

1. Introduction

Single image super-resolution (SR) aims to reconstruct a high-resolution (HR) image from a low-resolution (LR) observation, a problem that underpins applications from mobile photography to display rendering. Before deep learning, SR was often formulated with image priors such as sparse representation and internal self-similarity [1,2]. Since the seminal deep model SRCNN [3], accuracy has improved through deeper, recurrent, progressive, and feature-reuse networks, including VDSR [4], DRCN [5], LapSRN [6], DBPN [7], RDN [8], RCAN [9], and EDSR [10]. A recent survey summarizes this progression across architectures, losses, datasets, and metrics [11]. The corresponding growth in parameters and computation, however, makes many high-capacity models impractical for mobile and edge devices, where memory, latency, and energy are tightly constrained. This tension has

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motivated a sustained line of *lightweight* SR, which seeks a favorable trade-off between reconstruction quality and model cost.

Lightweight CNN-based SR.

Efficient CNN designs reduce cost by moving computation to low-resolution features, reusing parameters, simplifying feature distillation, or replacing heavy operators with compact mixing. FSRCNN accelerates SRCNN-style reconstruction [12], ESPCN performs efficient sub-pixel upsampling [13], and recursive or memory-based designs reuse features across depth [14,15]. CARN employs cascading residual connections with group convolutions [16]; IMDN introduces information multi-distillation [17]; RFDN reformulates this idea with residual feature distillation blocks [18]; and RLFN further simplifies local feature extraction for efficient SR [19]. More recent lightweight designs also explore shuffle-based convolutional mixing and dual-regression constraints [20,21]. Efficient-SR challenges reflect the same deployment pressure by ranking models with accuracy, runtime, parameters, and FLOPs considered together [22]. For the present work, the main limitation of these methods is not efficiency but context: locally connected convolutional operators cannot directly compare spatially distant yet semantically similar regions without stacking many layers.

Lightweight Transformer-based SR.

Long-range modeling tackles the context limitation from a different direction. Non-local operators aggregate features across distant positions [23], and Transformer-based restoration extends this idea with learned attention. IPT shows that large pre-trained Transformers can serve image restoration tasks including SR [24], while Swin Transformer provides the shifted-window operator that later restoration models adopt for tractable attention [25]. SwinIR introduces window-based attention for image restoration [26]; ELAN streamlines long-range attention to cut redundant computation [27]; SRFormer revisits the window–channel trade-off with permuted self-attention [28]; and HAT activates more pixels to enlarge the effective receptive field [29]. Efficient variants further mix local operators with omni-dimensional, n-gram, or hierarchical attention patterns [30–33]. ATD augments local windows with an adaptive token dictionary to extend the effective context [34]. While effective, window attention still ties token interaction to a fixed spatial partition. Semantically similar but spatially distant tokens therefore interact only when the window grows or an additional global path is introduced.

Content-aware / cluster-based routing.

A complementary direction groups tokens by *content* rather than by spatial position, allowing long-range but semantically related regions to interact within a compact attention. This idea is related to earlier self-exemplar and non-local SR designs, where similar patches or texture correspondences are explicitly mined across an image or reference [2,35,36]. SPIN performs superpixel-based token interaction for lightweight SR [37]. CATANet pushes this idea further with a Token Aggregation Block (TAB) that maintains a small set of routing centers, assigns each token to its nearest center, reorders tokens by cluster label, and aggregates the reordered sequence with an intra/inter-group attention operator [38]. This content routing delivers strong accuracy at a low budget and is the most direct predecessor of our work.

Limitations of center-based routing.

Despite its effectiveness, the TAB routing mechanism has four limitations. (i) Its routing centers are maintained across batches through a k-means–style iteration with an exponential moving average (EMA) buffer, so the centers are driven by a cross-batch historical state rather than strictly by the current image and adapt only slowly to per-image semantics. (ii) The token-to-center relation is a hard `argmax` label that discards the *confidence* of the assignment, treating a token that clearly belongs to one cluster identically to an ambiguous one on a boundary. (iii) The reordering clusters tokens only by label, with no intra-cluster purity constraint, so high- and low-confidence tokens are interleaved within a group during aggregation. (iv) Prototype *generation* and *confirmation* are entangled in a single iterative update, which weakens interpretability and stability. These observations suggest that a routing module built directly from the current input, that preserves and exploits assignment confidence, and that separates prototype generation from confirmation could improve content routing without sacrificing the lightweight regime.

Our approach.

We propose *Dynamic Prototype Routing* (DPR), a drop-in replacement for the TAB routing that keeps the CATANet backbone and its external tensor interface unchanged while remaining in a comparable lightweight operating regime. DPR generates semantic prototypes directly from the current tokens via a soft assignment, confirms them with a learnable query bank decoupled from generation, matches tokens back to the confirmed prototypes with a learnable temperature, and propagates an explicit per-token confidence through ordering and aggregation. The resulting network, **DPRNet**, is trained on DIV2K and evaluated on five standard benchmarks at $\times 2/\times 3/\times 4$, where it maintains CATANet-level reconstruction quality with small gains on most benchmark cells and a comparable lightweight cost profile, while exposing an interpretable, confidence-aware routing mechanism.

Contributions.

The main contributions are as follows.

- **C1 – Dynamic semantic prototypes.** We generate routing prototypes from the current tokens by a soft, assignment-weighted aggregation, removing the cross-batch EMA center buffer so that prototypes follow per-image semantics (Sec. 2.3, Eqs. (4)–(5)).
- **C2 – Decoupled prototype confirmation.** We separate prototype *generation* from *confirmation* by refining the content prototypes with a learnable query bank through a single gated cross-attention, improving the stability and interpretability of the routing slots (Eq. (7)).
- **C3 – Confidence-aware routing.** We explicitly retain the token assignment confidence and use it in three places: a composite ordering key that sorts high-confidence tokens first within each cluster, a confidence-modulated global gate, and a soft prototype fallback for uncertain tokens (Eqs. (10), (11), (12)).
- **C4 – Stabilized routing.** We introduce a learnable routing temperature together with a prototype usage-balancing regularizer to mitigate collapse onto a few prototypes and keep slot usage spread out (Eqs. (8), (13)).

Each design element is exposed as an independent switch that defaults to the full model and reduces to its disabled form when turned off, so the contributions can be isolated cleanly in the ablation study (Sec. 3.5).

2. Method

2.1. Overview

We build on the lightweight super-resolution (SR) backbone of CATANet and replace its token-routing component with a *Dynamic Prototype Routing* (DPR) module (Fig. 1). Given a low-resolution (LR) input $I_{LR} \in \mathbb{R}^{3\times H\times W}$, a 3×3 convolution extracts shallow features $F_0 \in \mathbb{R}^{C\times H\times W}$ (C=40). The deep feature extractor is a stack of L=8 blocks; each block couples a content-routed global aggregation unit (the Token Aggregation Block, TAB) with a local self-attention unit (LRSA), followed by a 3×3 fusion convolution and a residual connection:

$$F_l = F_{l-1} + \mathrm{Conv}_{3\times3}\big(\mathrm{LRSA}_l(\mathrm{TAB}_l(F_{l-1}))\big), \quad l = 1, \ldots, L. \tag{1}$$

A global residual links the deep features to the network output, which is reconstructed by a pixel-shuffle upsampler and added to a bilinearly upsampled base image:

$$I_{SR} = \mathrm{Up}\big(F_0 + F_L\big) + \mathrm{Bilinear}_{\uparrow s}(I_{LR}), \tag{2}$$

where $s \in \{2, 3, 4\}$ is the scale factor. The contribution of this work is entirely inside TAB: we keep its external interface $\mathbb{R}^{C\times H\times W} \to \mathbb{R}^{C\times H\times W}$ unchanged and replace the original means-based center iteration and hard-label sorting with the DPR pipeline (Sec. 2.3), while reusing the subsequent intra/inter-group aggregation operator (IASA, Sec. 2.4). This keeps the rest of the backbone untouched and isolates the effect of the routing mechanism.

Inside a TAB, the input feature map is flattened into a token sequence $X \in \mathbb{R}^{B\times N\times C}$ with $N{=}HW$, layer-normalized, routed by DPR, aggregated by IASA, and fused by a 1×1 convolution and a convolutional feed-forward network (ConvFFN), each with a residual connection. The data flow is

$$X \ \rightarrow\ \mathrm{DPR} \ \rightarrow\ \mathrm{IASA} \ \rightarrow\ \mathrm{Conv}_{1\times1} \ \rightarrow\ \mathrm{ConvFFN}. \tag{3}$$

2.2. Motivation: limitations of center-based routing

The original TAB performs content-aware reordering by maintaining a set of routing centers via a k-means–style iteration with an exponential moving average (EMA) buffer, assigning each token to its nearest center by an `argmax` hard label, sorting tokens by that label, and aggregating the sorted sequence. This design motivates DPR for four reasons. (i) The centers are driven by a cross-batch historical buffer rather than strictly by the current input, so they adapt slowly to per-image semantics. (ii) The token-to-center relation is a hard label and discards assignment confidence. (iii) The ordering clusters by label only, with no intra-cluster purity constraint. (iv) Prototype generation and prototype confirmation are entangled, which weakens interpretability. DPR responds by generating prototypes from the current tokens, decoupling generation from confirmation, and propagating an explicit assignment confidence through routing and aggregation.

2.3. Dynamic Prototype Routing

DPR maps a token sequence $X \in \mathbb{R}^{B\times N\times C}$ to a content-reordered sequence, M dynamic prototypes, and routing metadata. The pipeline contains four stages (Fig. 2).

(1) Dynamic prototype generation.

We first embed tokens with a lightweight head $E = \mathrm{GELU}(\mathrm{Linear}(\mathrm{LN}(X))) \in \mathbb{R}^{B\times N\times C}$, and predict a soft assignment of each token to M prototype slots:

$$A = \mathrm{softmax}\big(EW_a\big) \in \mathbb{R}^{B\times N\times M}, \quad W_a \in \mathbb{R}^{C\times M}, \tag{4}$$

where the softmax is taken over the M slots. Each prototype is the assignment-weighted average of the *current* tokens,

$$P^{c}_m = \frac{\sum_n A_{n,m} X_n}{\sum_n A_{n,m} + \epsilon}, \qquad P^{c} = \mathrm{normalize}\big(\mathrm{LN}(P^{c})\big), \tag{5}$$

with ℓ_2 normalization along the channel dimension. Unlike the EMA centers of the baseline, P^{c} depends only on the current input and carries no cross-batch state.

(2) Prototype confirmation by query refinement.

To separate prototype *generation* from *confirmation*, we refine P^{c} with a learnable query bank $Q \in \mathbb{R}^{M \times C}$ via a single cross-attention over the tokens. With query seed $\tilde{P} = P^{c} + Q$, queries from the seed, and keys/values from the embedded/raw tokens,

$$P^{r} = \mathrm{softmax}\Big(\frac{(\tilde{P}W_q)(EW_k)^{\top}}{\sqrt{d_r}}\Big)\,(XW_v), \tag{6}$$

where d_r is the router dimension. The refinement is fused through a learnable gate $\gamma = \sigma(g_r) \in (0,1)$,

$$P = \mathrm{normalize}\big(\mathrm{LN}(P^{c} + \gamma\, P^{r})\big). \tag{7}$$

Initializing g_r to a small value keeps the refinement a residual perturbation at the start of training, which helps keep prototype identities stable across samples. This stage can be disabled (ablation A1) by setting `use_prototype_query_refine`=false, which falls back to $P = P^{c}$.

(3) Confidence-aware token-to-prototype matching.

We match tokens *back* to the confirmed prototypes in a shared ℓ_2-normalized router space, $\hat{e}_n = \mathrm{normalize}(E_n W_t)$ and $\hat{p}_m = \mathrm{normalize}(P_m W_p)$, and compute routing scores with a *learnable temperature* τ:

$$S = \mathrm{softmax}\big(\tau\, \hat{E}\hat{P}^{\top}\big) \in \mathbb{R}^{B \times N \times M}, \qquad \tau = \min\big(e^{\theta},\, \tau_{\max}\big), \tag{8}$$

where θ is a learnable log-scale clamped by $\tau_{\max}$ for stability. A larger τ sharpens the assignment distribution and increases the separation of routing scores. For each token we read out the hard label and its confidence,

$$x^{s}_n = \max_m S_{n,m}, \qquad b_n = \arg\max_m S_{n,m}, \tag{9}$$

and *explicitly retain* the confidence x^{s}_n (addressing the hard-label confidence discarded by TAB).

(4) Confidence-aware ordering.

Instead of sorting only by the cluster label, we use a composite key that keeps the inter-cluster order while ordering high-confidence tokens first *within* each cluster:

$$k_n = b_n + 0.5\,(1 - x^{s}_n), \qquad \pi = \mathrm{argsort}_n(k_n). \tag{10}$$

Since $x^{s}_n \in (0,1)$, the additive term lies in $(0, 0.5)$ and never crosses an integer label boundary, so clusters stay contiguous while their members are ordered by confidence. The token sequence, labels, and confidences are gathered by π into X^{π}, b^{π}, $x^{s,\pi}$, and the inverse permutation π^{-1} is recorded for scatter-back. Setting `use_conf_sort`=false reduces Eq. (10) to a pure label sort $k_n = b_n$ (ablation A2).

DPR outputs the reordered tokens X^{π}, the dynamic prototypes P, the full score map S, and the per-token confidence x^{s}, which are consumed by the aggregator below.

2.4. Confidence-aware aggregation (IASA)

The aggregation operator is retained from the backbone but is made confidence-aware. On the reordered sequence X^{π} it computes (a) intra-group local attention over overlapping groups of size g, which captures fine local structure within each contiguous, semantically coherent segment; and (b) a global branch that attends to the dynamic prototypes P projected to keys/values. Let O_{loc} and O_{glb} denote the two branch outputs. They are combined through a learnable global gate $\beta = \sigma(g_g)$ modulated by the per-token confidence:

$$O = O_{\text{loc}} + \beta \cdot x^{s,\pi} \cdot O_{\text{glb}}, \tag{11}$$

so that low-confidence tokens rely less on the global prototype summary. The output is scattered back to the original token order via π^{-1} and projected. Setting `use_iasa_score_gate`=false removes the confidence factor in Eq. (11) ($O = O_{\text{loc}} + \beta\, O_{\text{glb}}$; ablation A2).

Soft prototype fallback.

To stabilize tokens whose hard assignment is uncertain, DPR adds a soft, confidence-weighted prototype context to the aggregated output. With a soft summary $\bar{X} = SP$ projected by W_s and a learnable gate $\alpha = \sigma(g_s)$,

$$Y = O + \alpha \left(1 - x^s\right) \left(\bar{X} W_s\right), \tag{12}$$

where $(1 - x^s)$ emphasizes low-confidence tokens. The result Y passes through the 1×1 convolution, residual connection, and ConvFFN. Setting `use_soft_fallback`=false reduces Eq. (12) to $Y = O$ (ablation A2). Equations (10), (11) and (12) together form the confidence-aware routing path, and the three switches are independent and default to the full model.

2.5. Prototype usage balancing

A learnable temperature can concentrate routing on a few prototypes. We therefore add a usage-entropy regularizer that encourages balanced slot use. Let $u_m = \frac{1}{N} \sum_n S_{n,m}$ be the mean usage of prototype m; its normalized entropy and the auxiliary loss are

$$\mathcal{H}(u) = -\frac{1}{\log M} \sum_m u_m \log u_m, \qquad \mathcal{L}_{\text{bal}} = \lambda \left(1 - \mathcal{H}(u)\right), \tag{13}$$

which is maximized (loss minimized) when usage is uniform. The total objective adds this term, summed over all blocks, to the pixel reconstruction loss:

$$\mathcal{L} = \mathcal{L}_{\text{pix}} + \sum_l \mathcal{L}_{\text{bal}}^{(l)}. \tag{14}$$

Setting the per-block weight λ=0 disables balancing (ablation A3).

2.6. Discussion of design

DPR keeps the backbone interface and parameter budget in the lightweight regime while changing only how tokens are organized before aggregation. The four stages map directly to the contributions: dynamic generation (Eqs. 4–5) for C1, generation/confirmation decoupling (Eq. 7) for C2, confidence-aware routing (Eqs. 8–10, 11–12) for C3, and the learnable temperature with usage balancing (Eqs. 8, 13) for C4. Because each design element is exposed as an independent switch that defaults to the full model and reduces to its disabled form when turned off, the contributions can be isolated cleanly in the ablation study (Sec. 3.5).

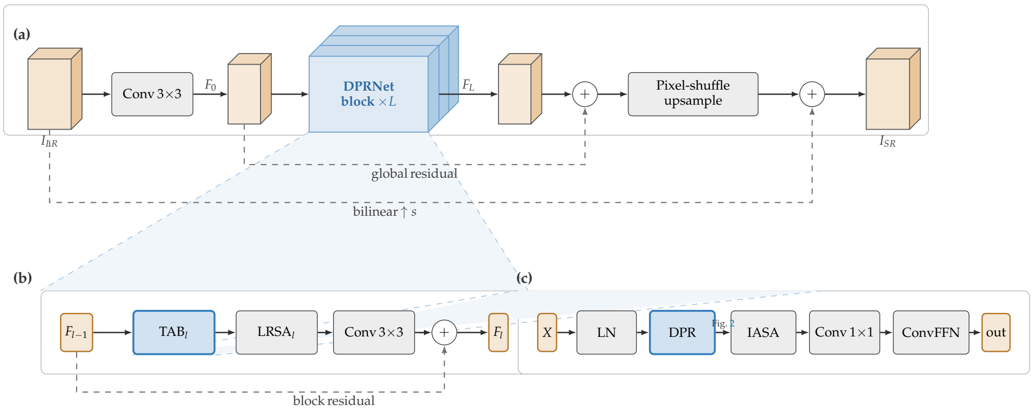

Figure 1. Overall architecture of DPRNet, following the lightweight-SR backbone of CATANet. **(a)** A 3×3 convolution extracts shallow features F_0, which pass through a stack of L=8 DPRNet blocks; the deep features are added back through a global residual and reconstructed by a pixel-shuffle upsampler that is summed with a bilinearly upsampled base image. **(b)** Each block keeps the TAB–LRSA–fusion structure under a residual connection. **(c)** Inside a TAB, the only modified component is the routing module: DPR replaces center-based routing while preserving the input/output interface, and is expanded in Fig. 2.

3. Experiments

3.1. Datasets and implementation details

Datasets and metrics.

Following the standard lightweight-SR protocol, we train on the 800 training images of DIV2K [39] and evaluate on five benchmarks: Set5 [40], Set14 [41], BSD100 [42], Urban100 [43], and Manga109 [44]. Low-resolution inputs are produced by bicubic downscaling at ×2, ×3, and ×4. We report PSNR and SSIM [45] on the Y channel of the YCbCr space, discarding a border of s pixels for scale s, which matches the convention used by the cited baselines.

Network configuration.

DPRNet keeps the CATANet backbone unchanged and replaces only the routing inside each TAB with the DPR module (Sec. 2). The deep feature extractor stacks L=8 blocks with channel width C=40. The number of prototype slots per block is $M = [16, 32, 64, 128, 16, 32, 64, 128]$, and the intra-group attention uses a group size of 128. The learnable routing temperature is initialized at τ=6 (i.e. a log-scale parameter $\theta = \log 6$) and clamped at $\tau_{\max}$=10 (Eq. (8)); the per-block usage-balancing weights are $\lambda = [5\times10^{-4}]_{\times 6}$ for the first six blocks and 7×10^{-4} for the last two (Eq. (13)). These yield 0.60–0.67 M parameters depending on scale (Table 2).

Training.

We first train the ×2 model from scratch for 800k iterations, and then finetune the ×3 and ×4 models for 250k iterations each from the ×2 checkpoint. Training patches are 128×128, 192×192, and 256×256 HR crops for $\times 2/\times 3/\times 4$ respectively, with a batch size of 16 and random horizontal flips and 90° rotations for augmentation. We optimize the L1 reconstruction loss together with the usage-balancing regularizer (Eq. (13)) using Adam (β_1=0.9, β_2=0.99, no weight decay) at an initial learning rate of 2×10^{-4}, halved by a MultiStep schedule (milestones at $\{300, 500, 650, 700, 750\}$k for ×2 and $\{125, 200, 225, 237.5\}$k for the $\times 3/\times 4$ finetuning). The learnable routing temperature is trained with a 0.1× learning-rate multiplier for stability. Validation is run every 2.5k iterations; the unified reporting checkpoint per scale is selected by the best benchmark-averaged PSNR/SSIM

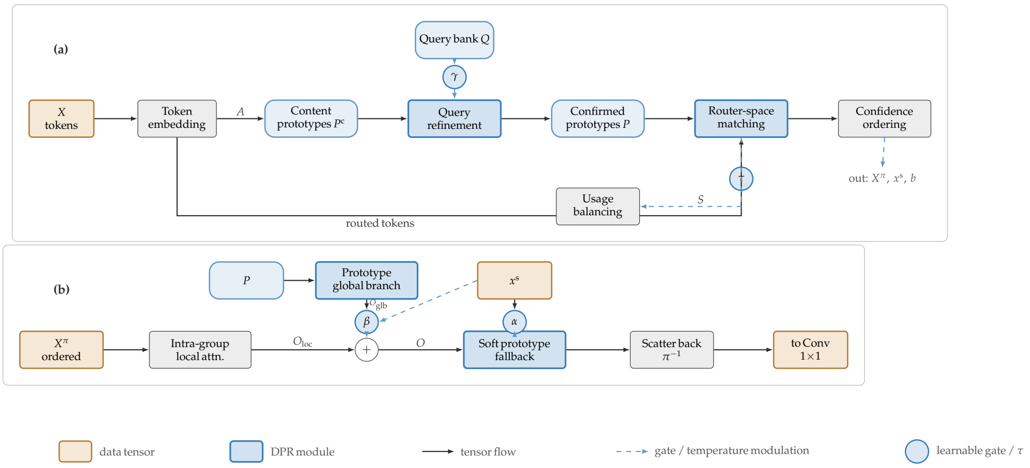

Figure 2. Dynamic Prototype Routing (DPR). **(a)** The router runs in a single pass: it embeds the tokens, forms content prototypes P^c from the current tokens, confirms them with a learnable query bank through a gated refinement (γ), and matches the tokens back to the confirmed prototypes P with a bounded temperature τ, yielding the ordered sequence X^π, the confidence x^s, and the label b (Eqs. (4)–(10)); the usage-balancing term acts on the score map S (Eq. (13)). **(b)** IASA consumes X^π, P, and x^s: a local branch and a prototype global branch are combined under a confidence-modulated gate β, and a soft prototype fallback gated by α stabilizes low-confidence tokens before the output is scattered back via π^{-1} (Eqs. (11)–(12)).

over Set5–Manga109, without per-dataset checkpoint selection. Experiments use PyTorch with the BasicSR framework on a single NVIDIA RTX 4090 GPU.

Efficiency measurement.

Parameters, Multi-Adds (MACs reported by thop), and inference latency are measured on a single NVIDIA RTX 4090 GPU for a fixed LR input of 256×256 (×4 output 1024×1024), averaged over 100 runs after 20 warm-up runs. This input convention matches the one under which the CATANet paper reports Multi-Adds for itself and the lightweight Transformer baselines, making the figures directly comparable. Efficient-SR challenge reports likewise treat runtime, parameters, and FLOPs as joint deployment criteria rather than auxiliary statistics [22]. To compare with FLOPs-based reports, the Multi-Adds values are multiplied by two.

3.2. Comparison with state-of-the-art

We compare DPRNet with eight representative lightweight SR methods. The CNN group contains CARN [16], IMDN [17], RFDN [18], and RLFN [19], while the Transformer and routing group contains SwinIR-light [26], ELAN-light [27], SRFormer-light [28], and the direct predecessor CATANet [38]. Competitor numbers are cited from the original papers, as noted in Table 1. DPRNet uses one unified checkpoint per scale and is reported on all five benchmarks without per-dataset selection.

Overall accuracy.

Table 1 shows that DPRNet and CATANet form the top tier at all three scales, with the clearest separation on Urban100 and Manga109 and smaller margins on Set5, Set14, and BSD100. For example, at ×4 DPRNet reaches 32.59 dB/0.9002 on Set5 and 27.76 dB/0.7434 on BSD100, compared with SRFormer-light at 32.51/0.8988 and 27.73/0.7422 under a comparable parameter budget. The pattern suggests that content-aware token aggregation is particularly useful in the lightweight regime when repeated or semantically related structures occur across distant positions.

Comparison with CATANet.

Against CATANet—the strongest baseline and the method from which DPRNet differs only in the routing module—DPRNet ranks first or tied first on nearly every benchmark/scale cell, with CATANet usually second. The margins are small but directionally consistent: at $\times 4$ DPRNet improves Urban100 from 26.87/0.8081 to 26.89/0.8089, Set5 from 32.58/0.8998 to 32.59/0.9002, Set14 from 28.90/0.7880 to 28.91/0.7883, and Manga109 PSNR from 31.31 to 31.32; at $\times 2$ it reaches the best Set14 (34.06/0.9220), Urban100 PSNR (33.15), and Manga109 (39.38/0.9789); and at $\times 3$ it leads or ties CATANet on every benchmark (e.g. BSD100 29.29/0.8103 and Urban100 29.06/0.8689). The single metric on which CATANet remains ahead is Manga109 SSIM at $\times 4$ (0.9183 vs. 0.9181). These gaps are within a few hundredths of a dB. The result therefore supports a conservative conclusion: replacing the EMA-center routing of TAB with the dynamic prototype routing of DPR preserves CATANet-level reconstruction quality while changing the routing mechanism to the interpretable, confidence-aware formulation analyzed in Sec. 3.5 and Sec. 3.6.

Efficiency.

Table 2 reports the $\times 4$ efficiency subset for methods whose Multi-Adds are available under the same fixed 256$\times$256 LR-input convention. DPRNet has 660K parameters and 52.14 G Multi-Adds, keeping it within the lightweight bracket. It uses fewer Multi-Adds than the lightweight Transformer baselines SwinIR-light (60.3 G) and SRFormer-light (56.5 G) while delivering competitive top-tier accuracy. Both its parameter count and Multi-Adds are modestly higher than CATANet's (660K/52.14 G vs. 535K/46.8 G at $\times 4$), reflecting the learnable query bank and router projections of DPR, so no efficiency advantage over CATANet is claimed. The per-scale DPRNet profile, including measured latency on our hardware, is given in Table 3. Cross-paper latency and compute values reported under incompatible conventions are excluded from Table 2 rather than mixed into the same numeric comparison.

3.3. Model complexity and efficiency

Table 2 compares model complexity at $\times 4$ for the subset of methods whose Multi-Adds are available under the same input convention, and Table 3 gives DPRNet's per-scale profile. All DPRNet numbers are measured under the protocol of Sec. 3.1 with a fixed 256$\times$256 LR input ($\times 4$ output 1024$\times$1024), reporting thop MACs and latency over 100 runs after 20 warm-ups. We use this 256$\times$256 input convention because it is the one under which the CATANet paper reports Multi-Adds for itself, SwinIR-light, and SRFormer-light; DPRNet's MACs are therefore directly comparable to those three cited figures. Methods whose published compute values use different conventions, or whose values are unavailable under this convention, are excluded from the efficiency subset rather than displayed as blank entries.

Parameters.

DPRNet stays within the lightweight bracket at 660K parameters ($\times 4$), comparable to Transformer baselines such as ELAN-light (601K) and well below SwinIR-light (897K), SRFormer-light (873K), and the CNN-heavy CARN (1592K). It is, however, larger than its direct predecessor CATANet (535K): the learnable query bank and the router/score projections of DPR add parameters relative to the EMA-center routing of CATANet. The comparison with CATANet is therefore framed at *matched-or-better accuracy* rather than as a parameter advantage (Urban100 26.89 vs. 26.87, with the full ablation analysis in Sec. 3.5). Fig. 3 plots this PSNR–parameter trade-off at $\times 4$ on Urban100: DPRNet attains

Table 1. Quantitative comparison with state-of-the-art lightweight SR methods on five benchmarks (PSNR(dB)/SSIM on the Y channel). **Best** and second-best results are highlighted. Competitor metrics are cited from the original papers (see notes). "-" denotes results not reported in the source.

Method	Params (K)	Set5 PSNR	Set5 SSIM	Set14 PSNR	Set14 SSIM	BSD100 PSNR	BSD100 SSIM	Urban100 PSNR	Urban100 SSIM	Manga109 PSNR	Manga109 SSIM
Scale ×2											
CARN	1592	37.76	0.9590	33.52	0.9166	32.09	0.8978	31.92	0.9256	38.36	0.9765
IMDN	694	38.00	0.9605	33.63	0.9177	32.19	0.8996	32.17	0.9283	38.88	0.9774
RFDN	534	38.05	0.9606	33.68	0.9184	32.16	0.8994	32.12	0.9278	38.88	0.9773
RLFN	527	38.07	0.9607	33.72	0.9187	32.22	0.9000	32.33	0.9299	-	-
SwinIR-light	878	38.14	0.9611	33.86	0.9206	32.31	0.9012	32.76	0.9340	39.12	0.9783
ELAN-light	582	38.17	0.9611	33.94	0.9207	32.30	0.9012	32.76	0.9340	39.11	0.9782
SRFormer-light	853	38.23	0.9613	33.94	0.9209	32.36	0.9019	32.91	0.9353	39.28	0.9785
CATANet	477	38.28	**0.9617**	33.99	0.9217	32.37	0.9023	33.09	**0.9372**	39.37	0.9784
DPRNet (ours)	602	**38.29**	**0.9617**	**34.06**	**0.9220**	**32.38**	**0.9031**	**33.15**	**0.9372**	**39.38**	**0.9789**
Scale ×3											
CARN	1592	34.29	0.9255	30.29	0.8407	29.06	0.8034	28.06	0.8493	33.43	0.9427
IMDN	703	34.36	0.9270	30.32	0.8417	29.09	0.8046	28.17	0.8519	33.61	0.9445
RFDN	541	34.41	0.9273	30.34	0.8420	29.09	0.8050	28.21	0.8525	33.67	0.9449
RLFN	-	-	-	-	-	-	-	-	-	-	-
SwinIR-light	886	34.62	0.9289	30.54	0.8463	29.20	0.8082	28.66	0.8624	33.98	0.9478
ELAN-light	590	34.64	0.9288	30.55	0.8463	29.21	0.8081	28.69	0.8624	34.00	0.9478
SRFormer-light	861	34.67	0.9296	30.57	0.8469	29.26	0.8099	28.81	0.8655	34.19	0.9489
CATANet	550	34.75	0.9300	**30.67**	**0.8481**	29.28	0.8101	29.04	**0.8689**	34.40	**0.9499**
DPRNet (ours)	674	**34.76**	**0.9301**	**30.67**	**0.8481**	**29.29**	**0.8103**	**29.06**	**0.8689**	**34.41**	**0.9499**
Scale ×4											
CARN	1592	32.13	0.8937	28.60	0.7806	27.58	0.7349	26.07	0.7837	30.42	0.9070
IMDN	715	32.21	0.8948	28.58	0.7811	27.56	0.7353	26.04	0.7838	30.45	0.9075
RFDN	550	32.24	0.8952	28.61	0.7819	27.57	0.7360	26.11	0.7858	30.58	0.9089
RLFN	543	32.24	0.8952	28.62	0.7813	27.60	0.7364	26.17	0.7877	-	-
SwinIR-light	897	32.44	0.8976	28.77	0.7858	27.69	0.7406	26.47	0.7980	30.92	0.9151
ELAN-light	601	32.43	0.8975	28.78	0.7858	27.69	0.7406	26.54	0.7982	30.92	0.9150
SRFormer-light	873	32.51	0.8988	28.82	0.7872	27.73	0.7422	26.67	0.8032	31.17	0.9165
CATANet	535	32.58	0.8998	28.90	0.7880	27.75	0.7427	26.87	0.8081	31.31	**0.9183**
DPRNet (ours)	660	**32.59**	**0.9002**	**28.91**	**0.7883**	**27.76**	**0.7434**	**26.89**	**0.8089**	**31.32**	0.9181

Notes. DPRNet (ours) uses a single unified checkpoint per scale (x2: `net_g_792500`; x3/x4: `net_g_250000`), reported across all benchmarks without per-dataset cherry-picking. Competitor numbers are cited verbatim: CARN/IMDN/RFDN/SwinIR-light/ELAN-light/CATANet from CATANet (CVPR 2025, Tab. 2); SRFormer-light from its paper (Tab. VI); RLFN from its paper (Tab. 1). RLFN does not report x3 or Manga109 in the original paper ("-"). Params for SwinIR-light and ELAN-light follow the convention of the CATANet table.

the highest PSNR among the compared lightweight methods while remaining within the same parameter bracket.

Computation.

In terms of Multi-Adds, DPRNet requires 52.14 G at ×4 under the unified 256×256 protocol. This is higher than CATANet's 46.8 G, so the learnable query bank and the router/score projections of DPR do add inference cost relative to CATANet's EMA-center routing. We therefore make no compute claim against CATANet, and the comparison stays at *matched-or-better accuracy* (Urban100 26.89 vs. 26.87, full ablation in Sec. 3.5). The compute profile is more favorable against the Transformer baselines: at the same input size DPRNet uses markedly fewer Multi-Adds than both SwinIR-light (60.3 G) and SRFormer-light (56.5 G), while reaching the best Urban100 PSNR in Table 2. DPRNet thus remains firmly inside the lightweight bracket, trading a modest cost increase over its direct predecessor for an interpretable, confidence-aware routing mechanism, and a clear cost reduction relative to attention-heavy Transformer designs at higher accuracy.

Table 2. Comparable efficiency subset on ×4 SR. Multi-Adds are MACs under a unified fixed LR input of 256 × 256 (×4 output 1024 × 1024); to compare against FLOPs-based reports, multiply by 2. Representative PSNR is on Urban100 (×4).

Method	Params (K)	Multi-Adds (G)	Urban100 PSNR	MAC source
SwinIR-light	897	60.3	26.47	CATANet Tab. 6
SRFormer-light	873	56.5	26.67	CATANet Tab. 6
CATANet	535	**46.8**	26.87	CATANet Tab. 6
DPRNet (ours)	660	52.14	**26.89**	measured

Notes. All Multi-Adds are MACs under a common 256 × 256 LR input (×4 output 1024 × 1024): DPRNet is measured under this protocol, and the CATANet, SwinIR-light and SRFormer-light values are taken from the CATANet paper (CVPR'25, Tab. 6), which uses the identical input convention, so the four entries are directly comparable. DPRNet needs more Multi-Adds than its direct predecessor CATANet (52.14 vs. 46.8 G) but markedly fewer than the Transformer baselines SwinIR-light (60.3 G) and SRFormer-light (56.5 G), while attaining the best Urban100 PSNR in this subset. Methods whose MACs are not reported under this shared input convention are omitted from the efficiency subset rather than filled from incompatible sources; the complete accuracy and parameter comparison remains in Table 1. Cross-paper latency is also omitted because runtime is hardware- and implementation-dependent.

Table 3. DPRNet per-scale efficiency profile (measured). All scales use a fixed LR input of 256 × 256; Multi-Adds are MACs (thop); latency averaged over 100 runs after 20 warm-ups on a single CUDA GPU.

Scale	LR Input	Params (K)	Multi-Adds (G)	Latency (ms)
×2	256×256	601.95	36.19	183.57 ± 7.70
×3	256×256	674.15	41.26	185.88 ± 6.33
×4	256×256	659.71	52.14	198.16 ± 9.43

Per-scale behavior.

Table 3 reports the profile at a fixed 256×256 LR input. Because the input size is held constant, the backbone cost is shared across scales and the Multi-Adds are dominated by the upsampling tail, so they *grow* with the scale factor—from 36.19 G (×2) to 41.26 G (×3) and 52.14 G (×4). Latency follows the same order (183.57, 185.88, 198.16 ms). The parameter count is essentially scale-independent (≈ 0.60–0.67 M), as only the input projection and the upsampling tail change with scale.

Scope of the efficiency comparison.

The main accuracy table (Table 1) remains the full comparison against all eight baselines. The efficiency table is deliberately narrower: it reports only common-protocol MACs plus DPRNet's measured per-scale runtime. We do not mix latency values across different devices or implementation stacks, and we do not import compute numbers reported under incompatible input-size or FLOPs-counting conventions. This makes Table 2 a directly comparable efficiency subset rather than an all-method table with non-comparable entries.

3.4. Qualitative comparison

Beyond aggregate PSNR/SSIM, Fig. 4 compares reconstructions at ×4 on three texture-dense hard samples selected by ground-truth high-frequency energy: two Urban100 images dominated by repetitive building grids (`img_092`, `img_024`) and one Manga109 page of high-contrast line art (`ThatsIzumiko`). These are exactly the regimes in which content-based

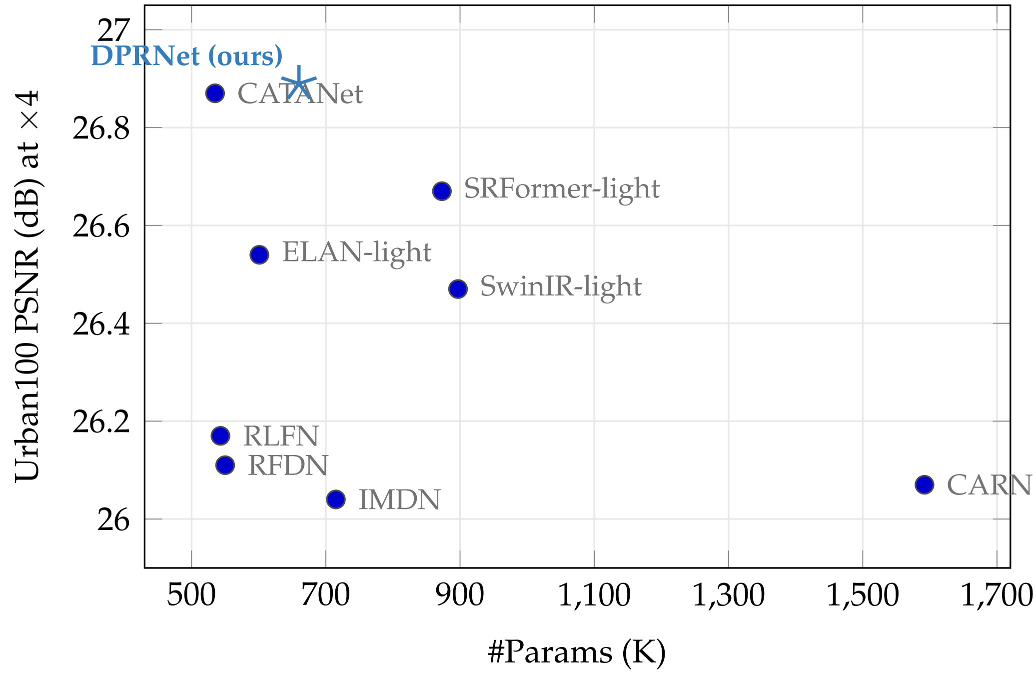

Figure 3. PSNR vs. parameter count at $\times 4$ on Urban100 (numbers from Table 2; competitor params cited from the original papers). DPRNet ($\star$) attains the highest PSNR among the compared lightweight methods while staying within the same parameter bracket (660K), trading a small parameter increase over its predecessor CATANet for the best reconstruction quality.

routing is expected to help most, since distant tokens that share the same structure can be aggregated together rather than only within a local window.

On the repetitive facades of Urban100, the bicubic baseline collapses adjacent window edges into blurred, aliased bands, and the lightweight CNN IMDN only partially recovers them. The transformer-based methods (SwinIR-light, SRFormer-light, CATANet) progressively sharpen the grid structure, and DPRNet reconstructs the window spacing with clean, well-separated boundaries on par with the strongest content-routing baseline, CATANet. On the Manga109 line art, where thin strokes are easily broken by upsampling, DPRNet keeps the contours continuous and avoids obvious ringing, matching the visual quality of the best transformer baselines. These observations are consistent with the routing diagnostics of Sec. 3.6: confidence-aware aggregation tracks structured, high-frequency regions that local-window methods resolve less faithfully. We stress that Fig. 4 is a qualitative comparison; the corresponding quantitative results for all baselines are reported in Table 1, where DPRNet remains in the top tier at a comparable parameter budget.

3.5. Ablation study

We ablate the four design elements of DPR using the independent switches exposed in Sec. 2. All ablations are conducted at scale $\times 4$, following CATANet [38] whose ablations are likewise performed at $\times 4$, the setting in which content routing over high-frequency texture is most stressed. To keep the compute budget tractable, each variant is *finetuned* from the converged full-model checkpoint and trained for a short 80k-iteration budget, with validation on Set5 and Urban100 (Y channel, matched `crop_border`) every 10k iterations; we report the iter-80k point. Besides PSNR/SSIM we log routing diagnostics per TAB block—the normalized prototype-usage entropy $\mathcal{H}(u)$ of Eq. (13), the active-prototype usage, and the learnable temperature—to inspect the routing behavior directly.

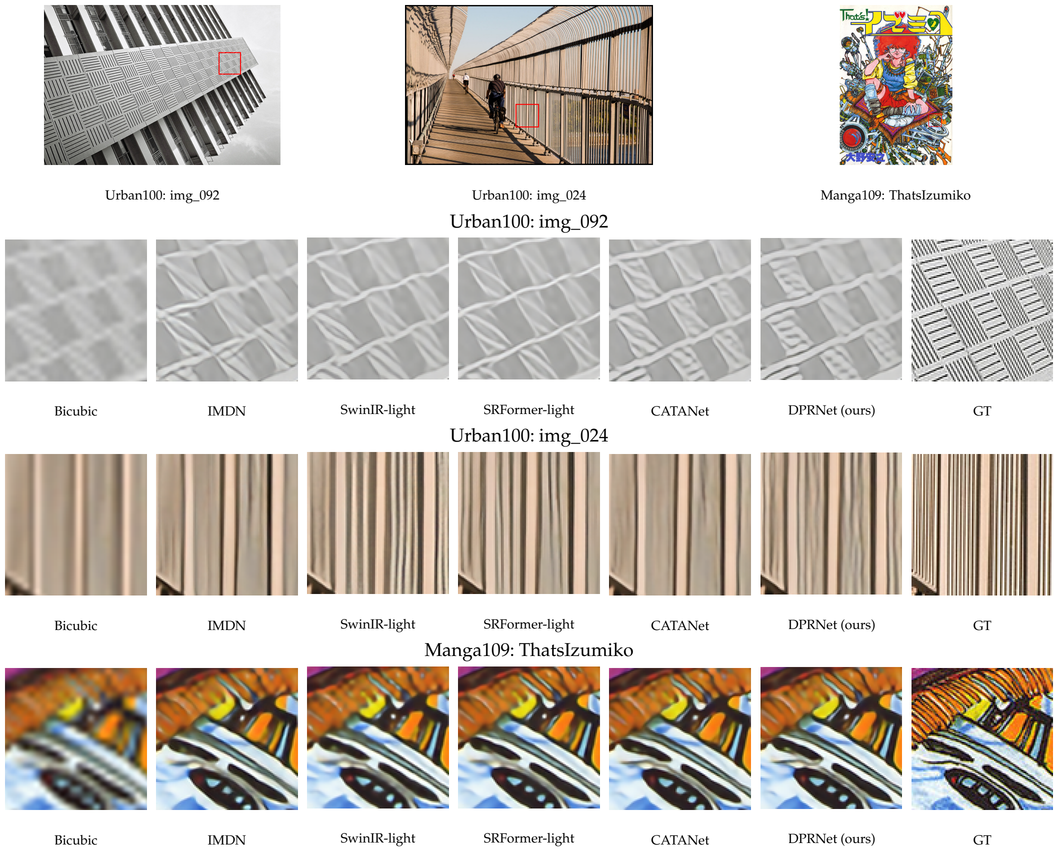

Figure 4. Qualitative comparison at $\times 4$ on texture-dense hard samples. The top reference strip shows the ground-truth images with the crop regions; the rows below magnify those regions for Bicubic, IMDN, SwinIR-light, SRFormer-light, CATANet, DPRNet (ours), and the ground truth. DPRNet restores repetitive building grids and high-contrast line art with clear structure, visually comparable to the strongest content-routing baseline on these challenging crops. The samples are chosen by high-frequency content rather than by per-crop DPRNet advantage, and quantitative PSNR/SSIM are reported in Table 1.

Protocol caveat.

Because every variant is finetuned from the *same* full-model initialization for a short budget, the disabled components are removed from an already converged representation rather than learned from scratch. Under this protocol the full model attains the best reconstruction in each ablation group, but the margins are small (up to $\sim$ 0.06–0.08 dB, against an $\sim$ 0.03 dB run-to-run fluctuation in the full model's own validation PSNR between adjacent 10k checkpoints) and the intermediate configurations are not strictly monotone in the number of enabled switches. We therefore read the PSNR/SSIM columns as evidence that each design element is *quality-neutral-to-mildly-positive and component-compatible*—the full configuration is never worse and is consistently the best, while no single switch can be credited with an isolated gain under this protocol—and we substantiate the routing-level effects through the diagnostics. A fully separated attribution of each switch would require retraining every variant from a switch-neutral initialization to the CATANet $\times 4$ from-scratch budget; this remains future work.

A1: decoupled prototype confirmation.

Table 4 compares DPR with and without the query-refinement stage (Eq. (7)). Enabling refine gives the better result on both benchmarks (Set5 32.59 vs. 32.55 dB; Urban100 26.89 vs. 26.84 dB), showing that the learnable query bank improves reconstruction under this protocol. As the refinement is gated by a small-initialized scalar (Eq. (7)), it acts as a residual confirmation step that keeps the prototype slots stable while decoupling generation from confirmation (C2); we therefore retain it by default. The motivation for dynamic, per-image prototypes over the cross-batch EMA centers of the baseline (C1) is supported indirectly by the published CATANet results rather than by this controlled switch, since re-implementing the EMA center branch is outside the scope of our ablation.

A2: confidence-aware routing.

Table 5 adds the three confidence-aware components cumulatively: confidence-aware ordering (Eq. (10)), the IASA score gate (Eq. (11)), and the soft prototype fallback (Eq. (12)). The full configuration (v4) is the best of the four on both Set5 and Urban100, but the intermediate steps are not monotone—adding the ordering and gate alone slightly lowers PSNR before the complete set recovers and surpasses the hard-routing baseline. Given the shared-initialization protocol above, this indicates that the three switches are mutually compatible and that the complete confidence-aware path is quality-preserving-to-mildly-positive, rather than each switch being individually accuracy-additive on top of a converged backbone. We keep the components because they make the routing path interpretable and confidence-driven: the retained per-token confidence x^{s} modulates ordering, the global gate, and the fallback (C3), which we visualize through the x^{s} distributions in Sec. 3.6.

A3: prototype usage balancing.

Table 6 isolates the usage-balancing regularizer (Eq. (13)). Enabling it raises the block-averaged normalized usage entropy from 0.6227 to 0.6865 on Urban100, and the increase is *consistent across all eight TAB blocks* (b_0–b_7; Table 7), i.e. the regularizer drives a more balanced use of prototype slots and counteracts the collapse that the learnable temperature would otherwise encourage (C4). The reconstruction quality is also slightly higher with the regularizer enabled (Set5 32.59 vs. 32.57 dB; Urban100 26.89 vs. 26.83 dB), so the balancing does not trade accuracy for a flatter usage distribution. The entropy effect is visible even under the short finetune protocol because it concerns the usage *distribution*, not absolute reconstruction quality. The complementary part of C4—a reduction of seed-to-seed variance—requires repeating the runs over multiple seeds; with a single seed available we report the balancing mechanism here and leave the variance analysis as future work, and we do not claim a variance reduction without those runs.

Summary.

The ablations show that DPR's design elements are quality-neutral-to-mildly-positive and mutually compatible under a shared-initialization finetune—the full model is the best variant in every group—and that the balancing regularizer produces a measurable, block-consistent increase in prototype-usage entropy. Stronger, fully separated accuracy attributions require from-scratch retraining; the present study supports the mechanism of each component without overstating its isolated contribution to PSNR.

3.6. Routing interpretability analysis

To inspect DPR's routing behavior independently of its effect on PSNR, we analyze the diagnostics logged during validation at $\times 4$ (iter 80k, Urban100). For each of the $L{=}8$ TAB blocks we record the per-token routing confidence $x^{\mathrm{s}} = \max_m S_{n,m}$ (Eq. (9)), the number of *active* prototypes (slots receiving at least one token), the normalized usage entropy $\mathcal{H}(u)$ (Eq. (13)), and the learned temperature τ (Eq. (8)). The blocks use $M = [16, 32, 64, 128, 16, 32, 64, 128]$ prototypes respectively.

Table 4. Ablation A1 (×4): prototype query refine. PSNR(dB)/SSIM (Y channel) at iter 80k and block-averaged normalized prototype-usage entropy on Urban100. Enabling refine gives the better reconstruction on both benchmarks under the short finetune protocol; it is retained as it improves (and does not degrade) quality while keeping slot assignment stable.

Variant	Set5		Urban100
	PSNR	SSIM	PSNR/SSIM
w/o refine	32.5494	0.8995	26.8358 / 0.8065
w/ refine (default)	32.5878	0.9001	26.8905 / 0.8086

Notes. ×4 ablation finetuned from the full-functioning checkpoint (`net_g_250000`) for an 80k budget; metrics read at iter 80k. Enabling refine yields the better PSNR/SSIM on both Set5 and Urban100, though all variants share this initialization and a short budget so the margin (∼0.04–0.05 dB) should not be over-read as a large isolated gain. The design motivation of dynamic prototypes vs. history-center+EMA is supported indirectly by the cited CATANet results (Table I), not by this controlled ablation.

Table 5. Ablation A2 (×4): incremental confidence-aware routing components. PSNR(dB)/SSIM (Y channel) at iter 80k. The full configuration (v4) attains the best reconstruction, but the intermediate steps are not monotone; under the short finetune protocol the components are mutually compatible and quality-preserving rather than individually accuracy-additive.

Variant	Switches			Set5		Urban100
	sort	gate	fb	PSNR	SSIM	PSNR/SSIM
v1 (hard sort)	✗	✗	✗	32.5459	0.8995	26.8559 / 0.8074
v2 (+conf. sort)	✓	✗	✗	32.5249	0.8995	26.8324 / 0.8065
v3 (+score gate)	✓	✓	✗	32.5116	0.8992	26.8123 / 0.8056
v4 (full, +soft fb)	✓	✓	✓	32.5878	0.9001	26.8905 / 0.8086

Notes. “sort”=confidence-aware sort key, “gate”=IASA score gating, “fb”=soft fallback. All variants are ×4 finetuned from the full-functioning checkpoint (`net_g_250000`) for an 80k budget; metrics read at iter 80k. The full configuration (v4) is the best of the four, but because the variants share this initialization and a short budget the intermediate PSNR/SSIM steps are not monotone; the components are reported as mutually compatible and quality-preserving rather than as a monotone incremental gain.

Balanced prototype usage (C4).

Fig. 6 reports the normalized usage entropy per block with and without the balancing regularizer. With balancing enabled (the full model) the entropy is higher in *all eight* blocks, raising the block-mean from 0.6227 to 0.6865; the effect is largest in the high-frequency-oriented block b_4 (0.7088 → 0.8466). Consistently, the average number of active prototypes increases from 47.5 to 52.3 across blocks, meaning that fewer slots are left unused. Because a learnable temperature can otherwise sharpen the assignment distribution and concentrate tokens on a handful of prototypes, these two measurements together show that the regularizer keeps prototype usage spread out and prevents slot collapse, matching the stabilizing role claimed for C4.

Bounded routing temperature (C4).

The learnable temperature is clamped by $\tau_{\max}=10$ for stability (Eq. (8)). In the trained full model it settles in a moderate range across blocks (per-block τ from 5.65 to 6.51, block-mean 6.06), staying well below the clamp rather than saturating it. The temperature therefore sharpens the routing scores enough to separate prototypes while the usage

Table 6. Ablation A3 (×4): prototype balance loss. Block-averaged normalized prototype-usage entropy on Urban100 (higher = more balanced usage) and reconstruction quality at iter 80k. The balance loss yields consistently higher usage entropy across all eight TAB blocks while also slightly improving reconstruction quality.

Variant	Usage entropy (mean, ↑)	Set5		Urban100
		PSNR	SSIM	PSNR/SSIM
w/o balance loss	0.6227	32.5689	0.8997	26.8262 / 0.8062
w/ balance loss (def.)	0.6865	32.5878	0.9001	26.8905 / 0.8086

Table 7. Per-block normalized prototype-usage entropy (Urban100, iter 80k). The balance loss increases usage entropy in every TAB block (b_0–b_7).

Variant	b_0	b_1	b_2	b_3	b_4	b_5	b_6	b_7
w/o balance	0.6482	0.5655	0.6947	0.6268	0.7088	0.5764	0.6279	0.5332
w/ balance	0.7181	0.5845	0.7227	0.6810	0.8466	0.6352	0.6848	0.6193

Notes. ×4 ablation finetuned from `net_g_250000` for an 80k budget; diagnostics read at iter 80k on Urban100. The balance loss consistently increases prototype-usage entropy (more balanced usage) across all blocks, supporting C4's balancing mechanism, and the PSNR/SSIM is slightly higher with it enabled. The seed-variance reduction sub-claim of C4 requires multi-seed runs (currently a single seed, 3407) and is therefore left outside the present claims.

regularizer keeps the assignment from collapsing—the two C4 components act as intended without hitting their safety bound.

Confidence-aware routing signal (C3).

The per-token confidence x^s that drives the ordering key (Eq. (10)), the IASA gate (Eq. (11)) and the soft fallback (Eq. (12)) is a non-degenerate signal in every block. Fig. 5 shows its distribution accumulated over *all* test images of Urban100 and Manga109 with the final model: in each block the mass lies clearly to the right of the uniform-assignment floor $1/M$ (for example, the Urban100 block means are 0.109 vs. 0.0625 for b_0 and 0.012 vs. 0.0078 for b_3) yet stays far from 1. The softmax assignment is therefore informative but soft rather than collapsed onto a single prototype, and its mode shifts toward the lower floor as M grows. Reshaping the original-order labels $b_n = \arg\max_m S_{n,m}$ to $H \times W$ yields the routing cluster maps in Fig. 7: the partitions track image content (repetitive building facades, high-contrast strokes) instead of a fixed spatial grid. In this visualization the important evidence is boundary continuity rather than the exact hue: early blocks form coarse content groups, while deeper larger-M blocks split the repeated structures into finer partitions. Both views are produced by a single forward pass with the cached routing state (`last_routing_map`, `last_x_scores`) and together confirm that the confidence used by the C3 components is meaningful.

Summary.

The logged diagnostics provide mechanism-level evidence for the routing design: the balancing regularizer produces a block-consistent increase in usage entropy and active-prototype count (C4), the learnable temperature stays in a stable, unsaturated range (C4), and the retained confidence signal is informative across all blocks (C3). These observations characterize the behavior of DPR independently of the reconstruction metrics discussed in Sec. 3.5.

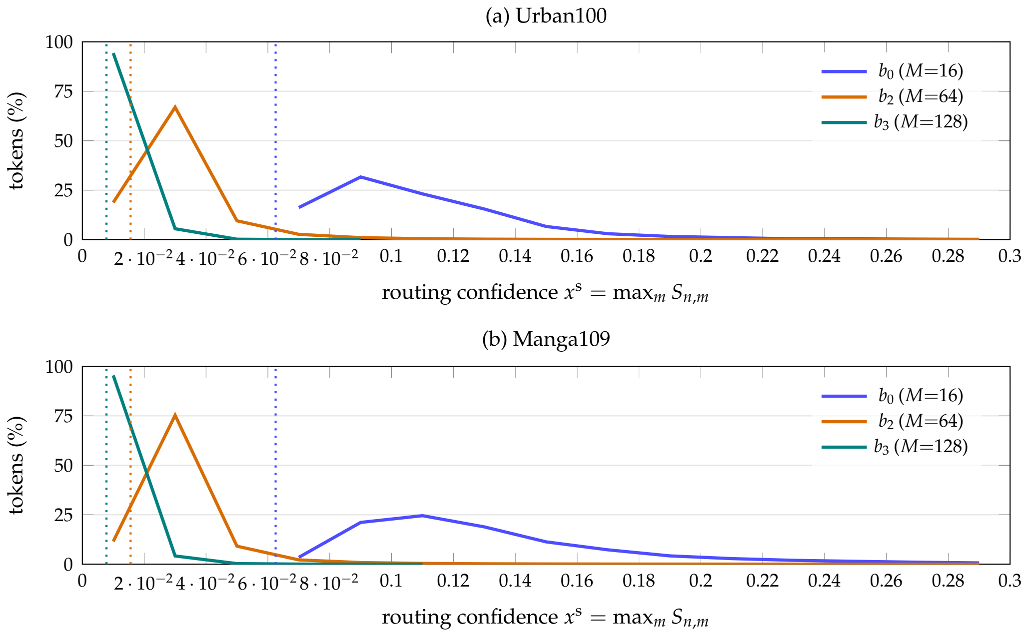

Figure 5. Distribution of the per-token routing confidence $x^{s} = \max_m S_{n,m}$ (Eq. (9)) at $\times 4$, accumulated over *all* test images of (a) Urban100 and (b) Manga109 with the final model. Curves are shown for three representative TAB blocks with increasing prototype count M; dotted vertical lines mark the uniform-assignment floor $1/M$. In every block the distribution sits clearly to the right of its floor (block-mean ≈ 1.5–$1.9 \times 1/M$) yet stays far from 1, confirming that the softmax assignment is *informative but soft*: tokens commit to a prototype more than chance while retaining the graded confidence that the C3 components (ordering key, IASA gate, soft fallback) rely on. As M grows the mode shifts toward the (lower) floor, consistent with confidence being spread over more slots.

4. Discussion

Content routing vs. spatial locality.

The benchmark results in Sec. 3.2 indicate where content-aware routing helps most. On Urban100 and Manga109—images dominated by repetitive structures and long-range self-similarity—DPRNet and CATANet, the two routing methods, separate more noticeably from the window- and distillation-based baselines, whereas on the more stochastic textures of BSD100 the margin between all top-tier methods narrows. This is consistent with the premise of content routing and with earlier SR work that exploits internal self-similarity or cross-scale non-local correspondences [2,35,43]: grouping tokens by semantics is most useful when semantically related content is spatially scattered, and less useful when local statistics already capture the signal. DPR does not discard spatial locality; it complements it, since each TAB is paired with a local self-attention unit (Sec. 2.1); the routing module supplies the long-range, content-based interaction that a fixed spatial window cannot.

Small margins and evidence strength.

The numerical margins over CATANet are deliberately framed as small. The value of DPR is therefore not a claim of large PSNR gain over its direct predecessor, but a combination of three evidence streams: a consistent top-tier position across five benchmarks and three scales (Sec. 3.2), a preserved lightweight profile relative to attention-heavy Transformer baselines (Sec. 3.3), and explicit routing diagnostics showing meaningful confidence scores and more balanced prototype usage (Sec. 3.6). This framing matters because modern lightweight SR methods are often separated by only a few hundredths of a dB under standard PSNR/SSIM evaluation; without mechanism-level analysis, such small differences would be weak evidence on their own.

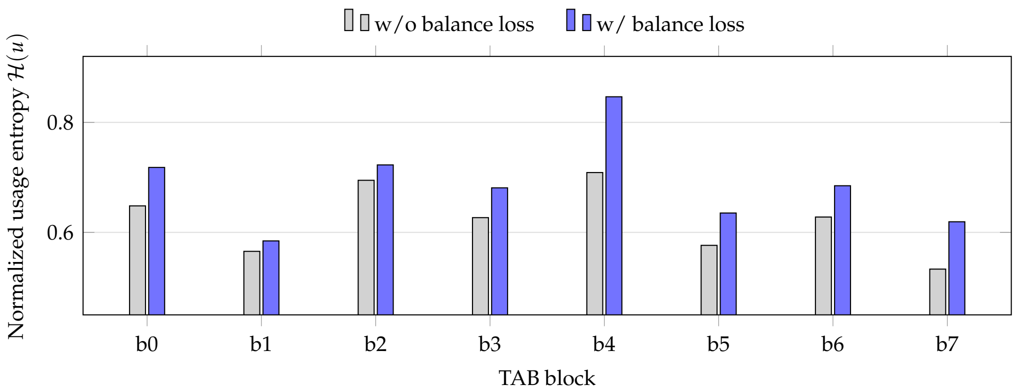

Figure 6. Normalized prototype-usage entropy $\mathcal{H}(u)$ (Eq. (13); higher = more balanced) per TAB block (b_0–b_7) at $\times 4$, iter 80k on Urban100. The usage-balancing regularizer increases the entropy in *all eight* blocks (block-mean $0.6227 \to 0.6865$), confirming that it counteracts the collapse onto a few prototypes encouraged by the learnable temperature (C4).

The risk of over-strong routing.

A confidence-aware router can, in principle, sharpen its assignments until tokens collapse onto a few prototypes, which would defeat the purpose of routing and destabilize training. Two components of DPR are explicitly designed against this failure mode: the routing temperature is clamped (Eq. (8)) and the usage-balancing regularizer (Eq. (13)) penalizes peaky usage distributions. The diagnostics in Sec. 3.6 show both mechanisms operating as intended—the learned temperature settles in a moderate, unsaturated range (block-mean 6.06, below the clamp of 10) and the balancing regularizer raises the prototype-usage entropy in all eight blocks ($0.6227 \to 0.6865$) while increasing the number of active prototypes. Thus, the benefits of confidence-aware routing are obtained without letting the router become degenerate.

Prototype count and the cost trade-off.

DPR uses a per-block prototype budget $M = [16, 32, 64, 128, 16, 32, 64, 128]$, growing within each stage. More prototypes give the router a finer content vocabulary but enlarge the query bank and the matching cost, while too few prototypes force unrelated tokens to share a slot. We keep the prototype schedule aligned with the CATANet backbone so that the comparison isolates the routing mechanism rather than a capacity change; this is also why DPRNet carries a modestly higher parameter count and Multi-Adds than CATANet (660K/52.14 G vs. 535K/46.8 G at $\times 4$ under a common 256×256 input), as the added cost lives in the query bank and projections of DPR. Even so, DPRNet stays within the lightweight bracket and below the Transformer baselines SwinIR-light (60.3 G) and SRFormer-light (56.5 G) at higher accuracy (Sec. 3.3). A systematic sweep of the prototype schedule is a natural extension but is orthogonal to the routing formulation studied here.

Limitations.

Several limitations bound the present study and frame our claims conservatively. First, the ablations (Sec. 3.5) finetune every variant from the same converged full-model checkpoint under a short 80k-iteration, single-seed budget; this isolates whether each design element is quality-neutral-to-mildly-positive and mutually compatible, but it cannot attribute an isolated PSNR gain to any single switch, and the seed-variance component of C4 is therefore left as future work rather than asserted. A fully separated attribution would require retraining each variant from a switch-neutral initialization at the CATANet $\times 4$ from-scratch budget. Second, the motivation for dynamic, per-image prototypes over cross-batch EMA centers (C1) is supported indirectly through the published CATANet results

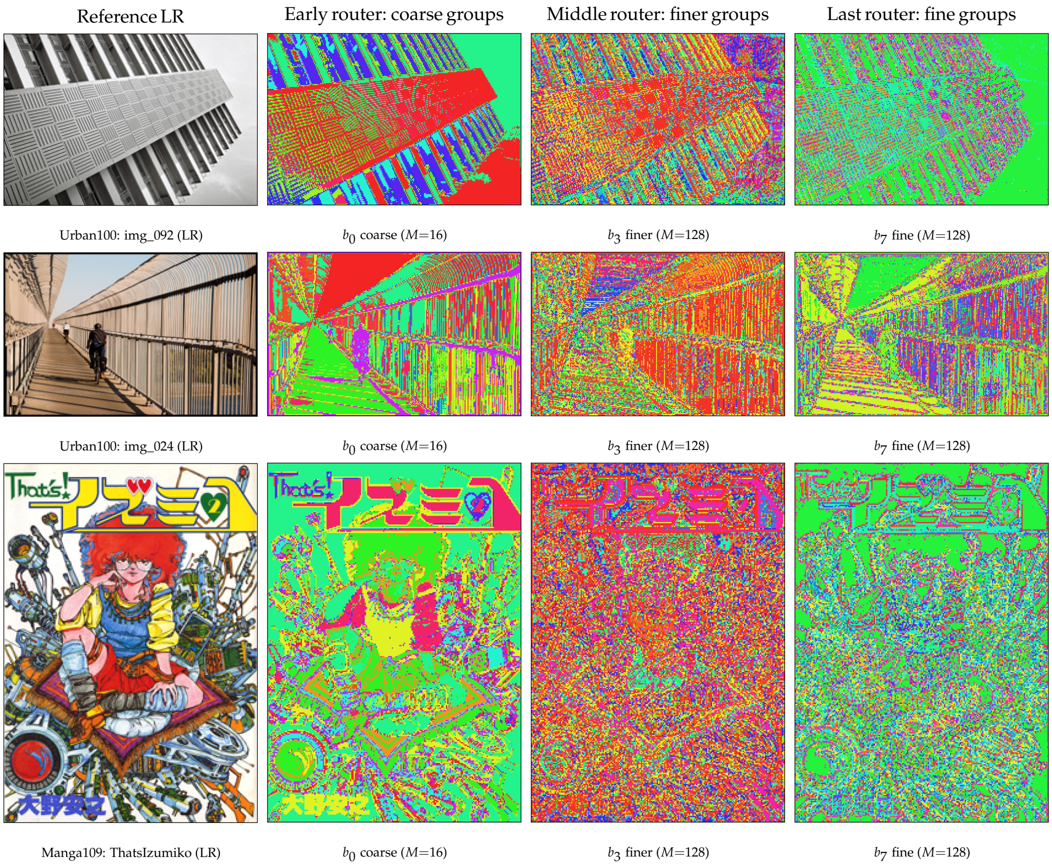

Figure 7. Routing cluster maps at ×4. Each LR pixel is colored by its assigned prototype index $b_n = \arg\max_m S_{n,m}$ (Eq. (9)) and reshaped from the original token order to $H \times W$. Read each row by comparing the cluster boundaries with the reference LR image: the early block forms coarse content groups, while the deeper blocks split repeated facade elements and high-contrast strokes into finer partitions. Colors encode prototype identity within a block and are not comparable across blocks, so the meaningful cue is region continuity rather than a specific hue. This visualizes the same routing signal whose confidence x^s is quantified in Fig. 5.

rather than through a re-implemented EMA branch. Third, the efficiency comparison is intentionally limited to common-protocol MACs and our measured DPRNet runtime: we do not claim cross-method latency because the competing models were not remeasured on the same hardware and implementation stack (Sec. 3.3). Finally, models are trained on DIV2K only; larger training sets (e.g. DF2K) and real-world or arbitrary-scale degradations are not explored here and may further shift the trade-off.

5. Conclusion

We presented Dynamic Prototype Routing (DPR), a drop-in replacement for the center-based Token Aggregation Block of CATANet that keeps the backbone and tensor interface unchanged while remaining in a comparable lightweight operating regime. DPR generates semantic prototypes directly from the current tokens, decouples their confirmation through a learnable query bank, matches tokens back with a learnable temperature, and propagates an explicit per-token confidence through ordering, gating, and a soft fallback, with a usage-balancing regularizer that keeps prototype use spread out. On five standard benchmarks at $\times 2/\times 3/\times 4$, the resulting DPRNet maintains CATANet-level reconstruction quality with small gains on most benchmark cells and a comparable lightweight cost profile. The routing diagnostics provide mechanism-level evidence that the temperature remains unsaturated and prototype usage becomes measurably more balanced across all blocks.

Because DPR preserves the external interface of the routing block, it is not tied to this backbone and can serve as a confidence-aware content-routing module for other token-aggregation architectures. Future work includes separating the contribution of each design element through from-scratch retraining from a switch-neutral initialization and a multi-seed variance study, remeasuring all comparison models for cross-method runtime under a single implementation protocol, and extending DPR to larger training sets and real-world or arbitrary-scale degradations.

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